

PRAKASH DHUNGANA

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PROFESSIONAL SUMMARY

Embedded software engineer with production experience developing and debugging safety-critical C/C++ software for autonomous driving and in-vehicle systems on embedded Linux, QNX, NVIDIA DriveOS, and RTOS platforms. Integrated NVIDIA DriveOS Safety Services into a production automotive middleware stack under ISO 26262, and built low-level PCIe and DMA drivers enabling deterministic communication on multicore SoCs. PhD research focused on performance optimization for embedded systems — spanning real-time latency, memory, and power constraints — with deep expertise in TinyML and edge intelligence, deploying and optimizing quantized ML workloads on resource-constrained edge devices.

CORE COMPETENCIES

Embedded C/C++ • Embedded Linux (NVIDIA Jetson/Xavier, DriveOS; Yocto, Buildroot, systemd) • Kernel & Device Driver Development (PCIe, DMA) • Multi-threading, Concurrency & Memory Management • RTOS & Real-Time Systems (QNX, Bare-metal) • Vehicle Networks (CAN, Automotive Ethernet, I2C, SPI, UART) • Debugging & Profiling (TRACE32/Lauterbach, GDB, QNX Momentics, Logic Analyzers) • ISO 26262 Functional Safety, AUTOSAR, MISRA-C/C++ • TinyML/Edge Intelligence • Python, Bash, Git, CI/CD, Docker

PROFESSIONAL EXPERIENCE

Graduate Research Assistant – Embedded AI & Real-Time Systems

Jan 2023 – Aug 2026

University of Kentucky, Lexington, KY, USA

- Conducted research on real-time and embedded systems optimization for resource-constrained platforms; published peer-reviewed research and presented findings at international conferences.
- Architected custom integer-only inference engines in C/C++ for ARM Cortex-M MCUs, eliminating floating-point overhead to minimize latency (41 ms) and memory footprint (13.5 kB).
- Interfaced bare-metal firmware with hardware to profile and optimize quantized ML model execution on constrained edge devices, applying disciplined memory management under tight resource budgets.
- Developed novel feature extraction pipelines using wavelet packets and custom C/C++ kernels, achieving optimal keyword-spotting accuracy with 14 ms feature extraction and 6 ms quantized inference.
- Implemented on-device continual learning algorithms enabling embedded models to adapt to domain shifts in real time, with and without cloud dependency (on-device/federated learning).

Embedded Software Engineer

Jul 2021 – Dec 2022

TTTech Auto (now TrustMotion), Osijek, Croatia

- Integrated NVIDIA DriveOS Safety Services (3LSS) into the MotionWise autonomous-driving middleware platform, strictly adhering to ISO 26262 ASIL safety concepts and requirements.
- Implemented low-level C/C++ drivers for PCIe and DMA on multicore SoCs, enabling deterministic inter-processor communication between safety and performance hosts.
- Debugged complex system-level issues across middleware and low-level software using Lauterbach TRACE32, analyzing software components, stack usage, and intercore communication in a real-time operating system environment.
- Configured Basic Software (BSW) modules including MCAL, NvM, and DEM using DaVinci Configurator and Generator, ensuring seamless integration with the Runtime Environment (RTE).
- Performed static code analysis and unit testing with VectorCAST, collaborating with QA on test strategy and ensuring compliance with MISRA-C/C++ standards.

Graduate Student Intern – ADAS Embedded Software

Nov 2019 – Jun 2021

TTTech Auto (now TrustMotion), Osijek, Croatia

- Optimized an automotive vision pipeline on TI TDA2x heterogeneous SoCs, reducing DSP execution time by 89.87% through frame-wise lookup tables.
- Implemented C/C++ image-processing pipelines for panoramic image correction, meeting strict 60 fps real-time latency constraints on automotive hardware.
- Designed and deployed a real-time panoramic image correction algorithm (color and brightness adjustment) optimized for the ALPHA board.

Automotive Service Engineer

Dec 2016 – Aug 2019

IME Motors / Goldfish International, Kathmandu, Nepal

- Diagnosed complex faults in engines and vehicle subsystems using onboard diagnostic tools and Diagnostic Trouble Codes (DTCs).
- Analyzed electrical schematics and powertrain control module logic to resolve hardware-software interface issues in commercial vehicles.

KEY PROJECTS

- **TinyML for Real-Time Audio Perception** (C/C++, Assembly) — Designed integer-only signal processing and classification pipelines for keyword spotting on MCUs; achieved real-time quantized inference under strict latency and memory constraints; implemented on-device continual learning for robustness under domain shifts.
- **Panoramic Image Correction on ADAS Platform** (C++, DSP, RTOS) — Proposed an alternative to color balance for brightness and color equalization; implemented and validated brightness equalization algorithms on real automotive ADAS hardware for panoramic images.

EDUCATION

Ph.D. in Computer Engineering , University of Kentucky, Lexington, KY, USA	2023 – 2026
M.Sc. in Automotive Computing and Communications , University of Osijek, Croatia	2019 – 2021
B.E. in Mechanical Engineering , Tribhuvan University, Kathmandu, Nepal	2012 – 2016

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Bash, Assembly

Operating Systems & RTOS: Embedded Linux, NVIDIA DriveOS, QNX, RTOS, Bare-metal

Embedded Linux: Buildroot, Yocto (OpenEmbedded), Cross-compilation Toolchains, Bootloaders, systemd

Embedded Platforms: NVIDIA Jetson (embedded Linux) & Xavier, ARM Cortex-M/A, Aurix TriCore

Driver & Low-Level Development: Kernel device driver development, PCIe, DMA, Ethernet NIC, USB, MIPI camera, FPGA interfaces, Board bring-up

Systems Programming: Multi-threading, Concurrency, Memory Management, System Architecture, Requirements Analysis, SDLC

Communication Protocols: CAN, Automotive Ethernet, TCP/IP, UDP, I2C, SPI, UART

Testing & Debugging: TRACE32 (JTAG/Lauterbach), GDB, QNX Momentics, Logic Analyzers, VectorCAST, Profiling, Unit/Integration/System-level testing

Automotive Standards: ISO 26262, AUTOSAR, ASPICE, MISRA

DevOps & Tools: Git, CI/CD pipelines, Docker, Jira, Bitbucket, Eclipse

Embedded AI / TinyML: TensorFlow Lite, Edge Inference, Integer-arithmetic Optimization, On-device Continual Learning

Autonomy & Robotics: Real-time Control, Sensor/Actuator Integration, ADAS Perception Pipelines, Computer Vision

CERTIFICATIONS

- Linux Device Drivers – Communicating with Hardware; Linux Kernel Programming
- Automotive Software with AUTOSAR; Automotive Software with MotionWise
- Advanced C++ Programming
- Deep Learning A-Z (Neural Networks, TensorFlow)
- Self-Driving Car Engineering (Deep Learning Applications for Autonomy)